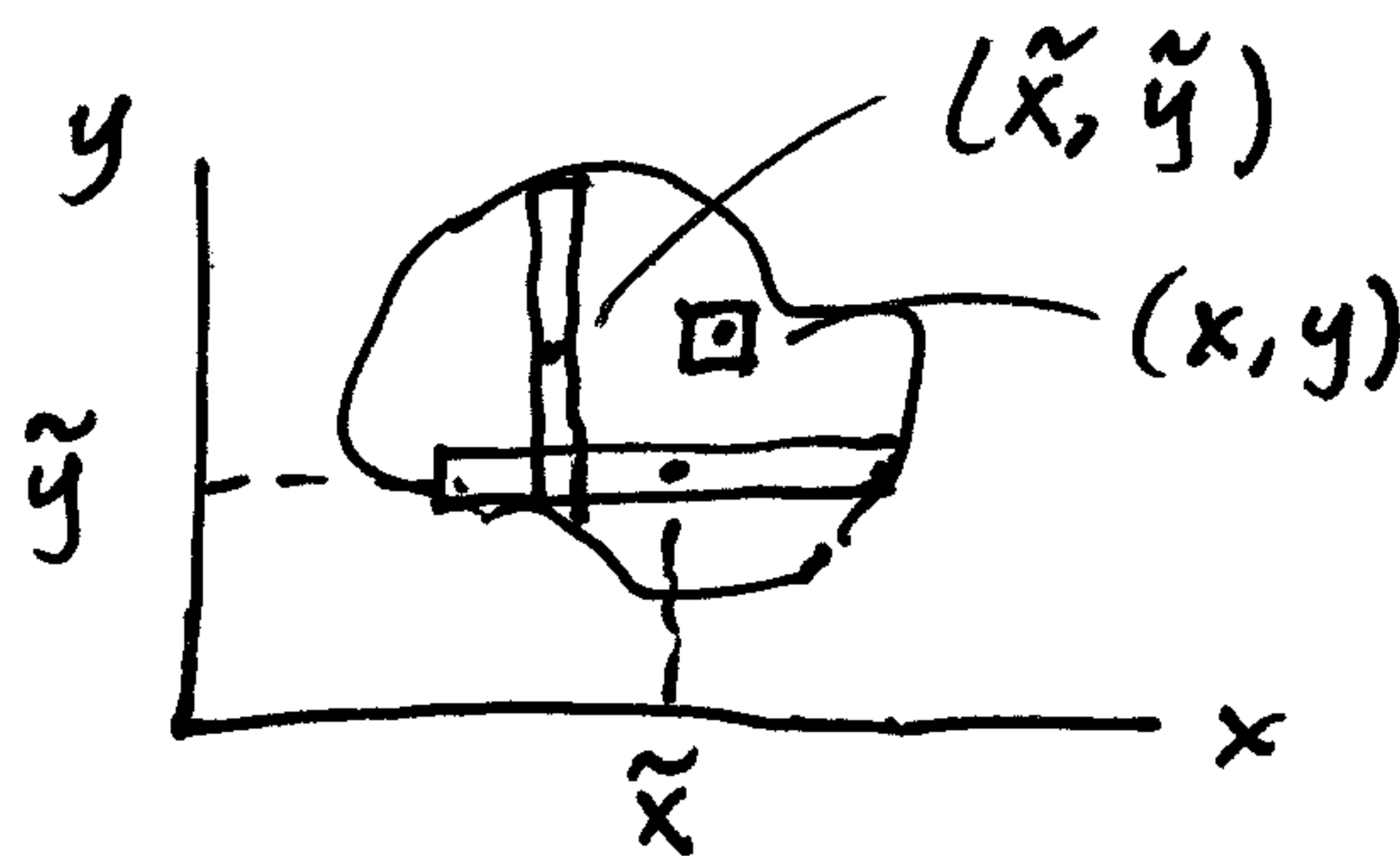


Hibbeler Crib Sheet

①

Area Centroid

$$\bar{x} = \frac{\int_A \tilde{y} dA}{\int_A dA}$$



where, $\tilde{x}$ = dist to centroid of dA
 $\tilde{y}$ = " " " " "

$$\bar{y} = \frac{\int_A \tilde{x} dA}{\int_A dA}$$

You can use any dA



as long as define $(\tilde{x}, \tilde{y})$ correctly.

BUT note on the Hibbeler crib sheet:

For Area MOI (Moment of Inertia)...

He defines these as:

$$I_x = \int_A y^2 dA$$

$$I_y = \int_A x^2 dA$$

He does not use
 $\tilde{x}$ or $\tilde{y}$!!

Why?

Short answer: You cannot use these definitions
for I_x and I_y with any dA like we
did for centroids!

Be careful!

Easiest dA 's to use for Area MOI's:

②

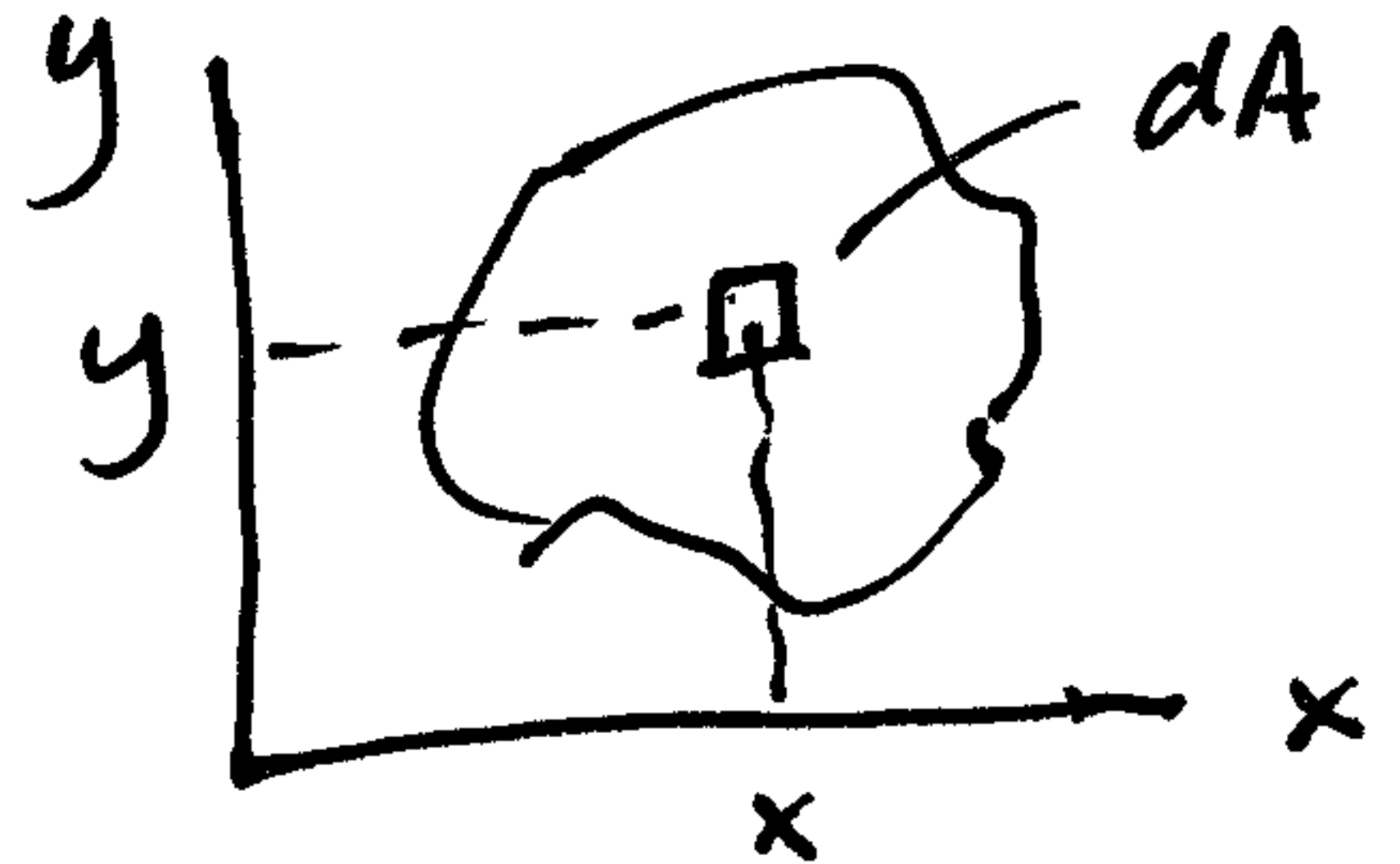
- ① Use $dA = dx dy$ or $dA = dy dx$
and double integral.

This is the easiest and best, in my opinion.

Then,

$$I_x = \int_A y^2 dA$$

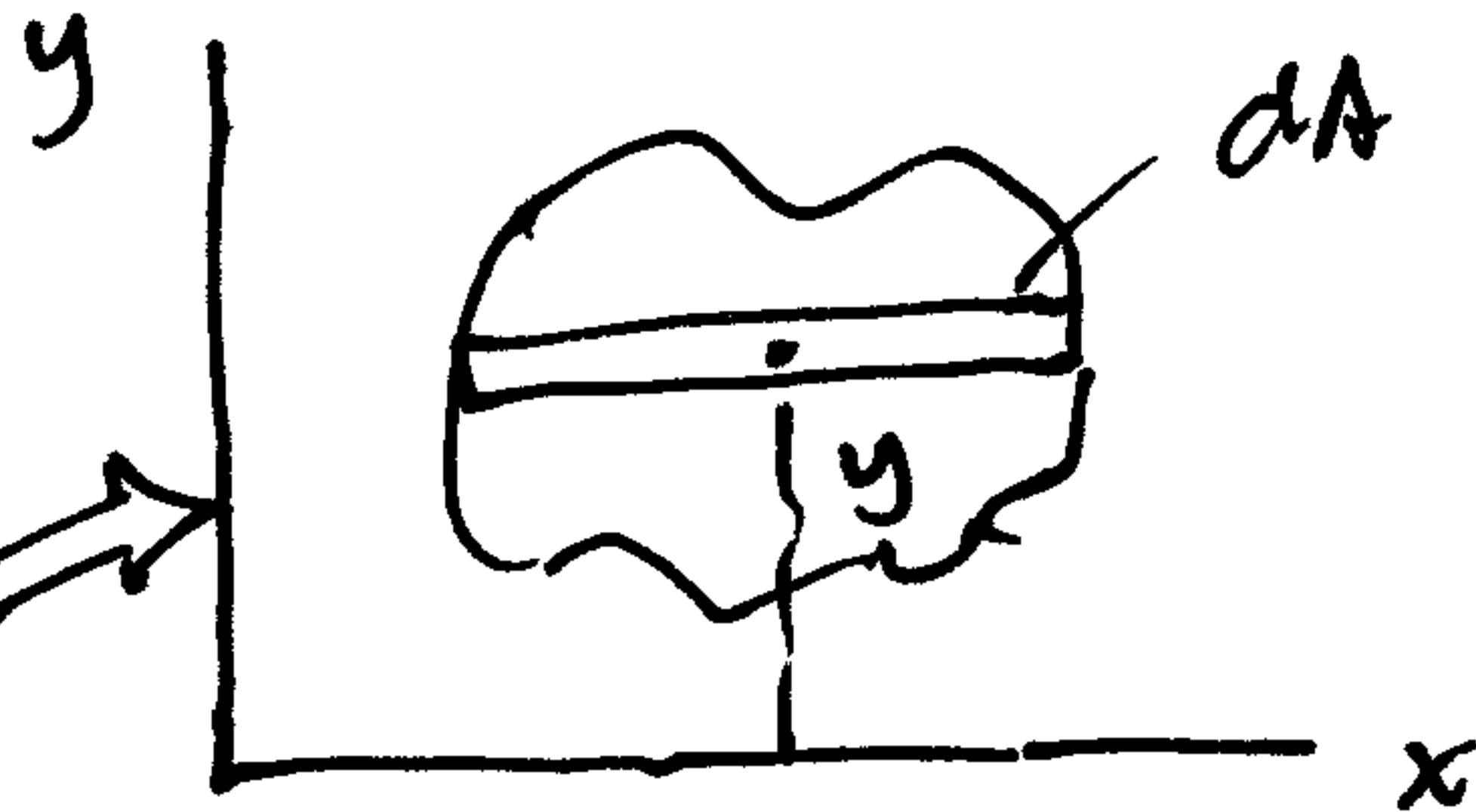
$$I_y = \int_A x^2 dA$$



- ② Another good choice is to use dA 's
parallel to the axes:

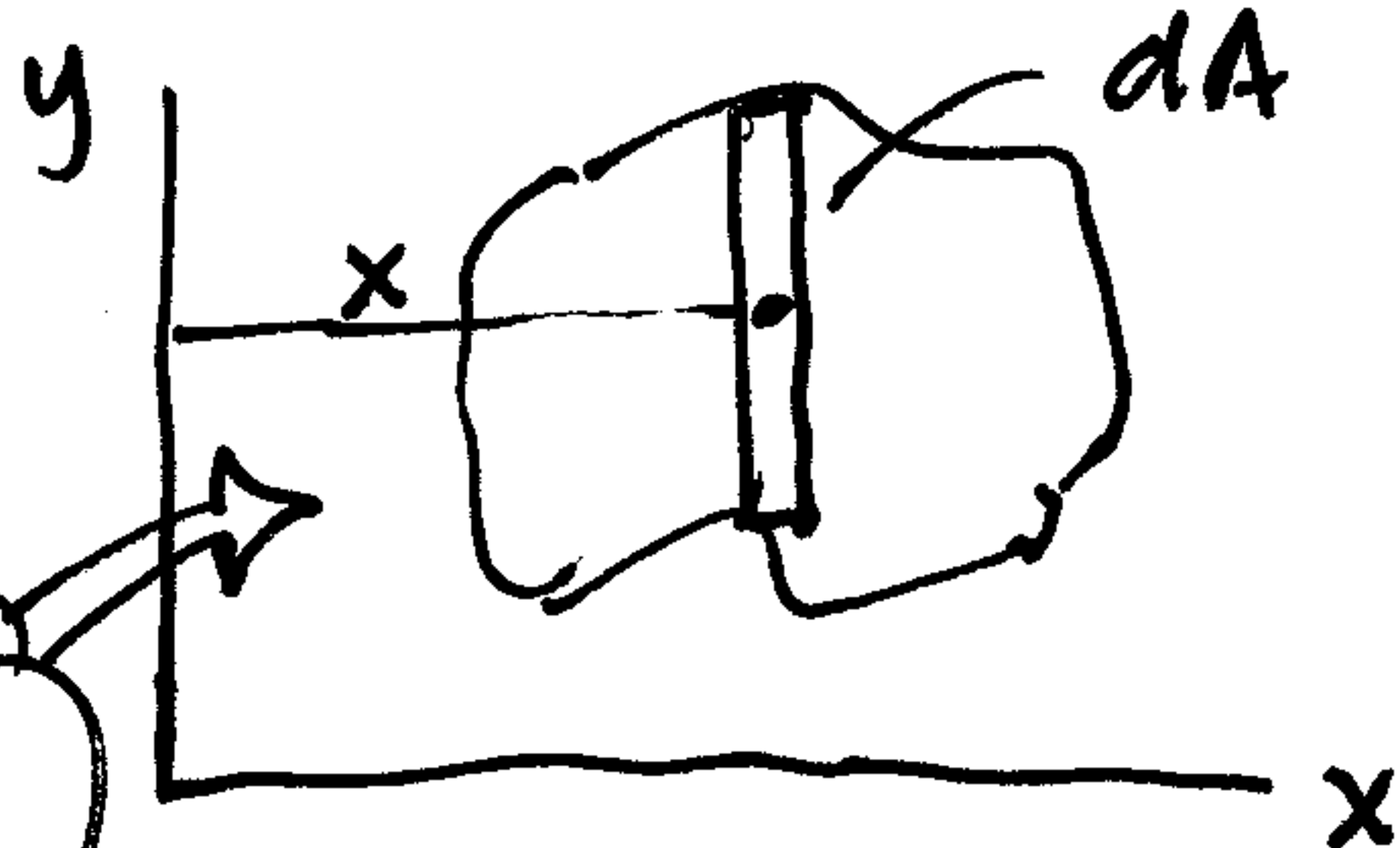
thus,

Entire element
is dist y
from x axis
and



$$I_x = \int_A y^2 dA$$

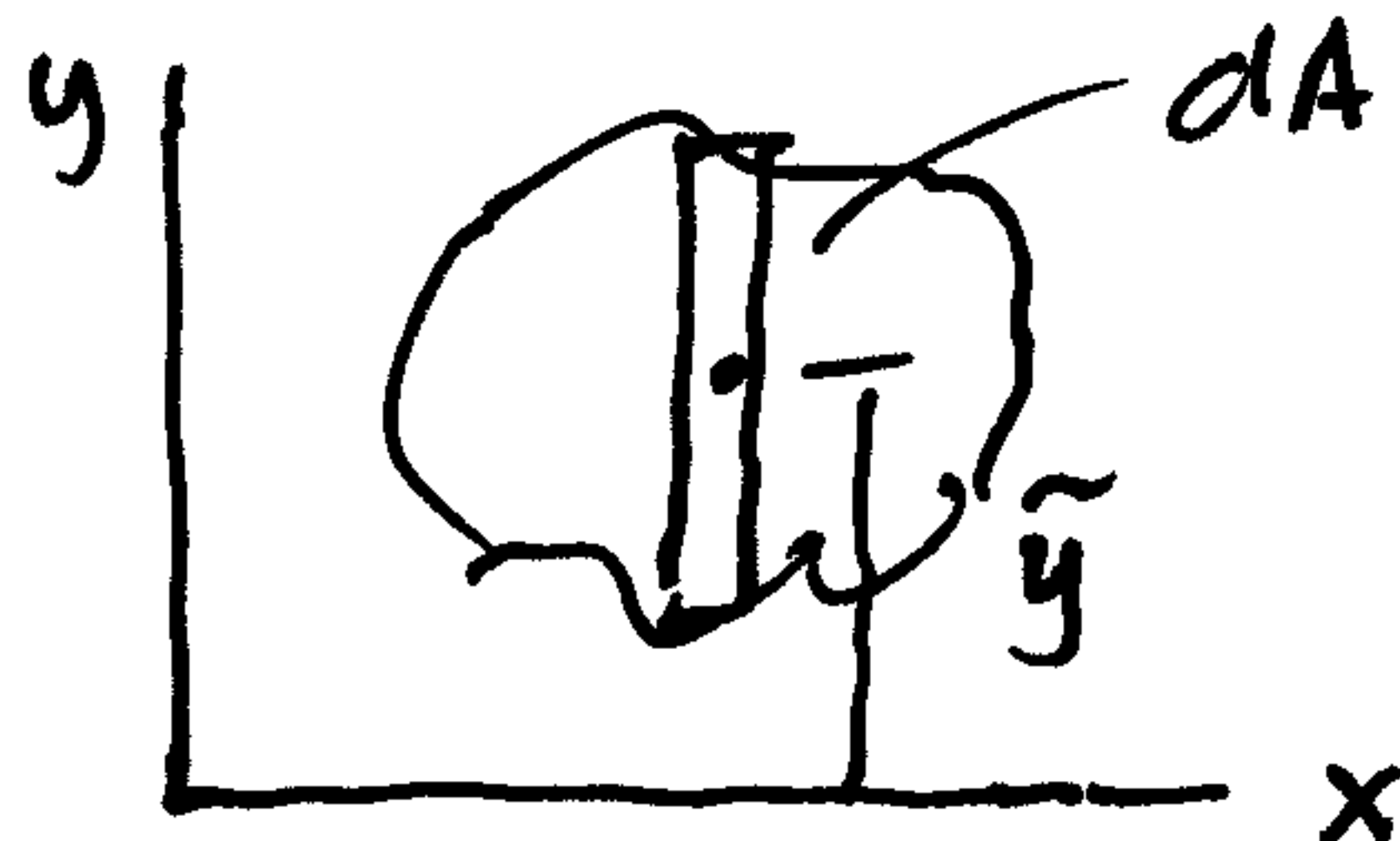
Entire element
is dist x
from y axis



$$I_y = \int_A x^2 dA$$

③ But can we use dA 's $\perp$ to the axis of interest? ③

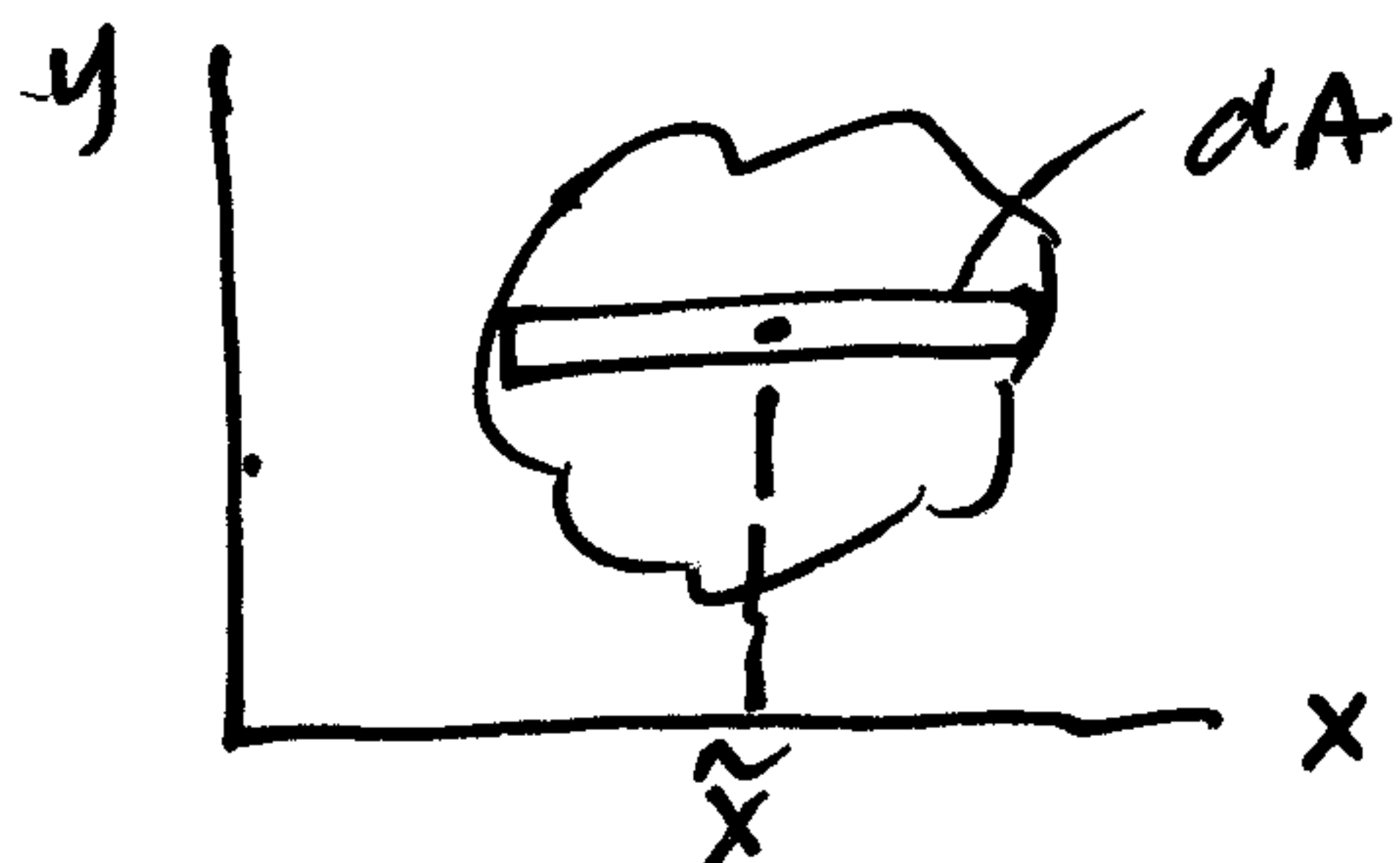
e.g.



$$I_x = \int_A \tilde{y}^2 dA \quad ?$$

NO! This is not valid!

or



$$I_y = \int_A \tilde{x}^2 dA \quad ?$$

NO! Not valid!

④ Why do these dA 's NOT work, but the others do?

Short answer:

$$I_x = \int_A y^2 dA$$

When we write

or

$$I_y = \int_A x^2 dA$$

* with the proper dA 's !!

we are summing the MOI's of the dA 's about the respective axes.

In other words, we are really writing...

$$I_x = \int dI_x, \quad \text{where } dI_x = \text{MOI of the } dA \text{ about the } x \text{ axis}$$

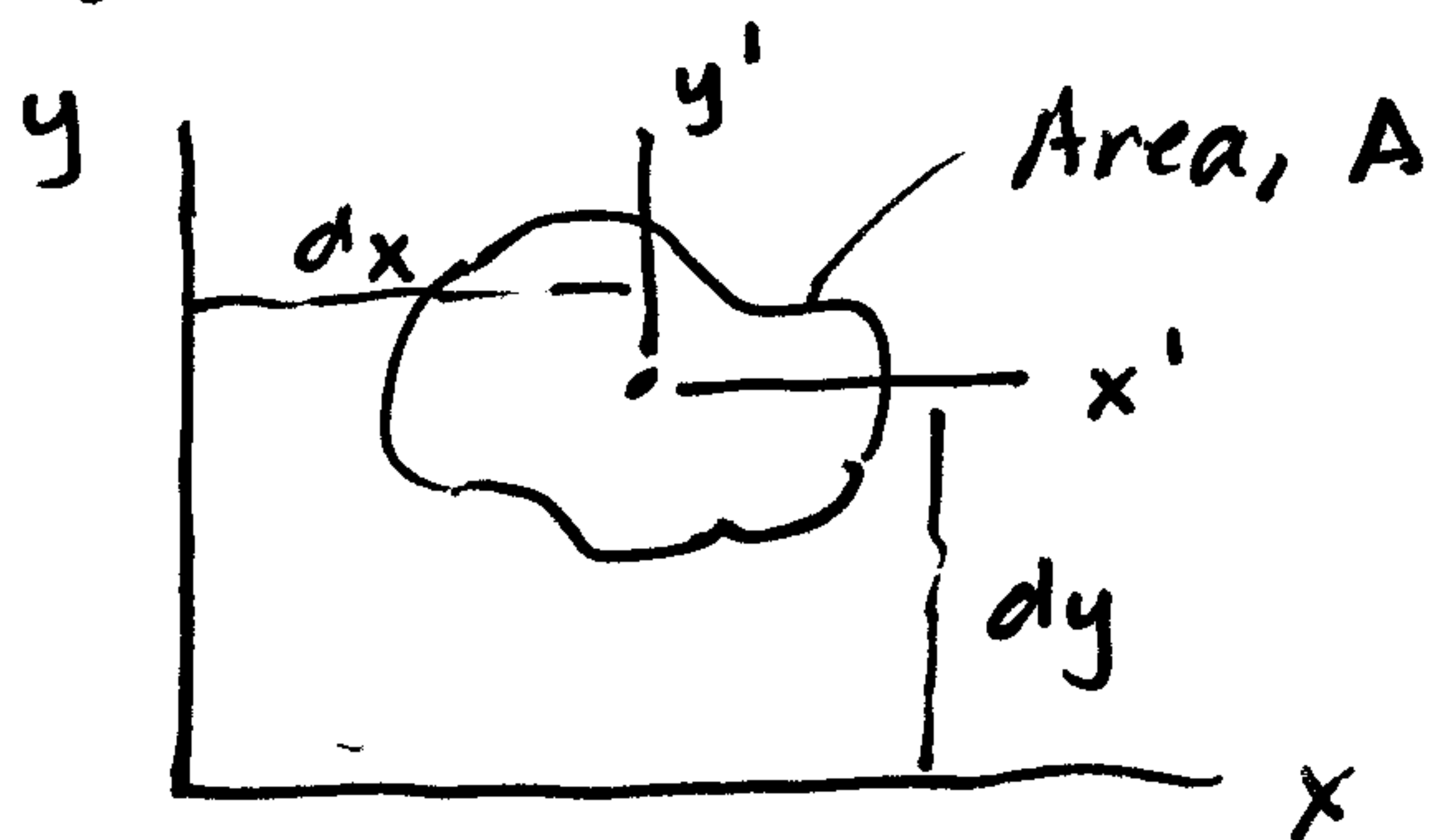
and

$$I_y = \int dI_y, \quad \text{where } dI_y = \text{MOI of the } dA \text{ about the } y \text{ axis}$$

⑤ Which leads us to the Parallel Axis Theorem (PAT)

④

(we'll derive the PAT soon...)



x' and y' are centroidal axes...

What is the PAT?

$$I_{Axis} = \bar{I} + Ad^2$$

It's used to "transfer" a MOI about an area's centroidal axis ($\bar{I}$) to another axis parallel to it.

Thus, for our area, A , above,

if we know $\bar{I}_{x'}$, A , and dy ,

then

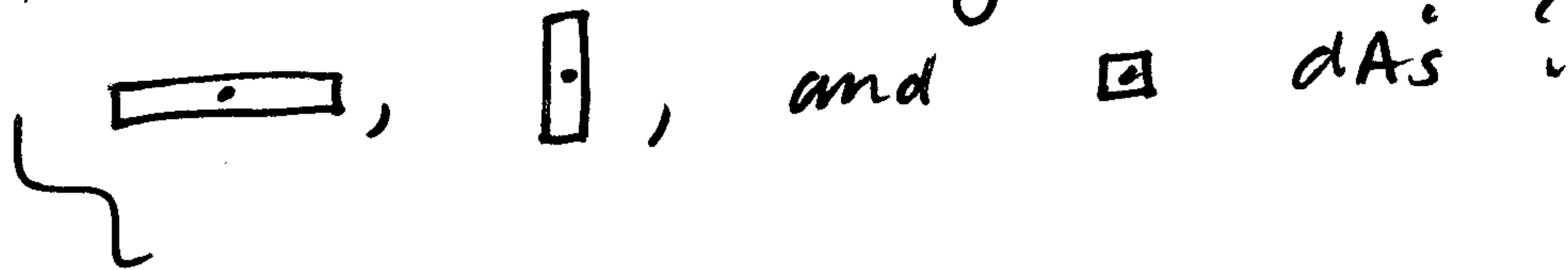
$$I_x = \bar{I}_{x'} + A \cdot dy^2$$

And, if we know $\bar{I}_{y'}$, A , and dx ,

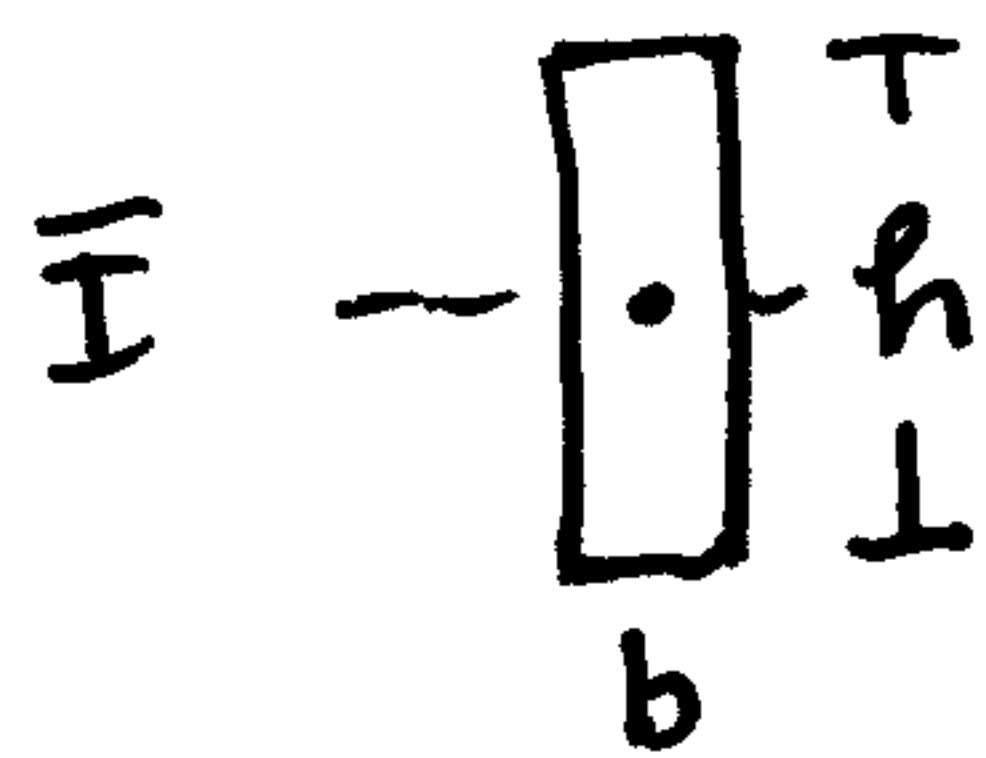
then

$$I_y = \bar{I}_{y'} + A \cdot dx^2$$

⑥ So how do these apply to



All of these dA 's are rectangles...

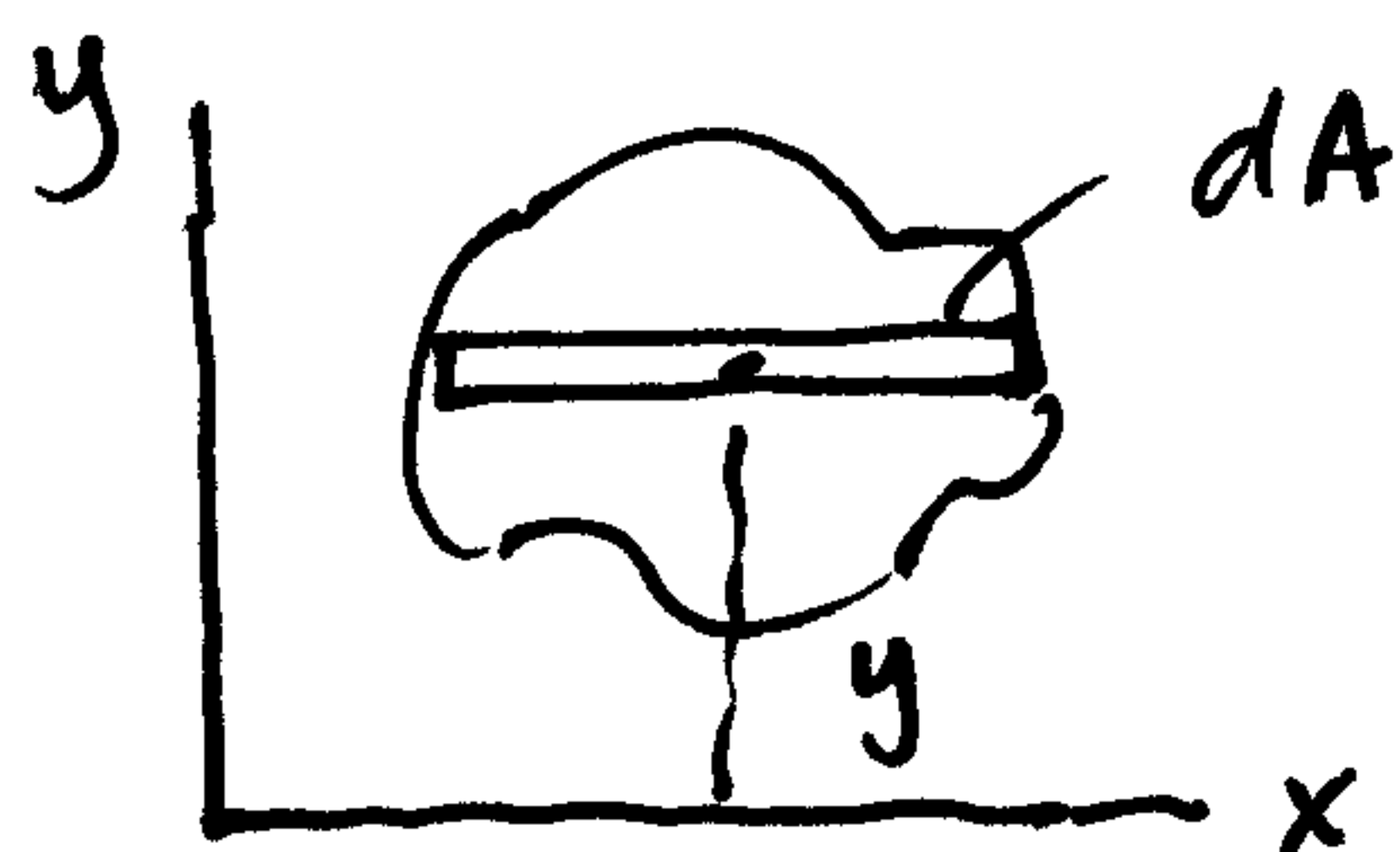


for a general rectangle about its centroidal axis,

its
$$\bar{I} = \frac{1}{12} b h^3$$

⑦ So how does this PAT apply to various dA 's?

⑤



If our most general formula for MOI is

$$I_x = \int dI_x$$

we can use the PAT in differential form to find I_x of the dA !

PAT in differential form:

$$dI_x = d\bar{I}_{x'} + dA \cdot y^2$$

and

$$dI_y = d\bar{I}_{y'} + dA \cdot x^2$$

For the case above,

A diagram of a rectangular differential element with width b and height $h = dy$. The width b is indicated by a horizontal arrow below the rectangle, and the height $h = dy$ is indicated by a vertical arrow to the right of the rectangle.

$$\text{thus, } d\bar{I}_{x'} = \frac{1}{12} b (dy)^3 = 0!$$

$$\text{thus, } dI_x = y^2 dA$$

finally,

$$I_x = \int dI_x = \int y^2 dA$$

for an element parallel to the x axis!

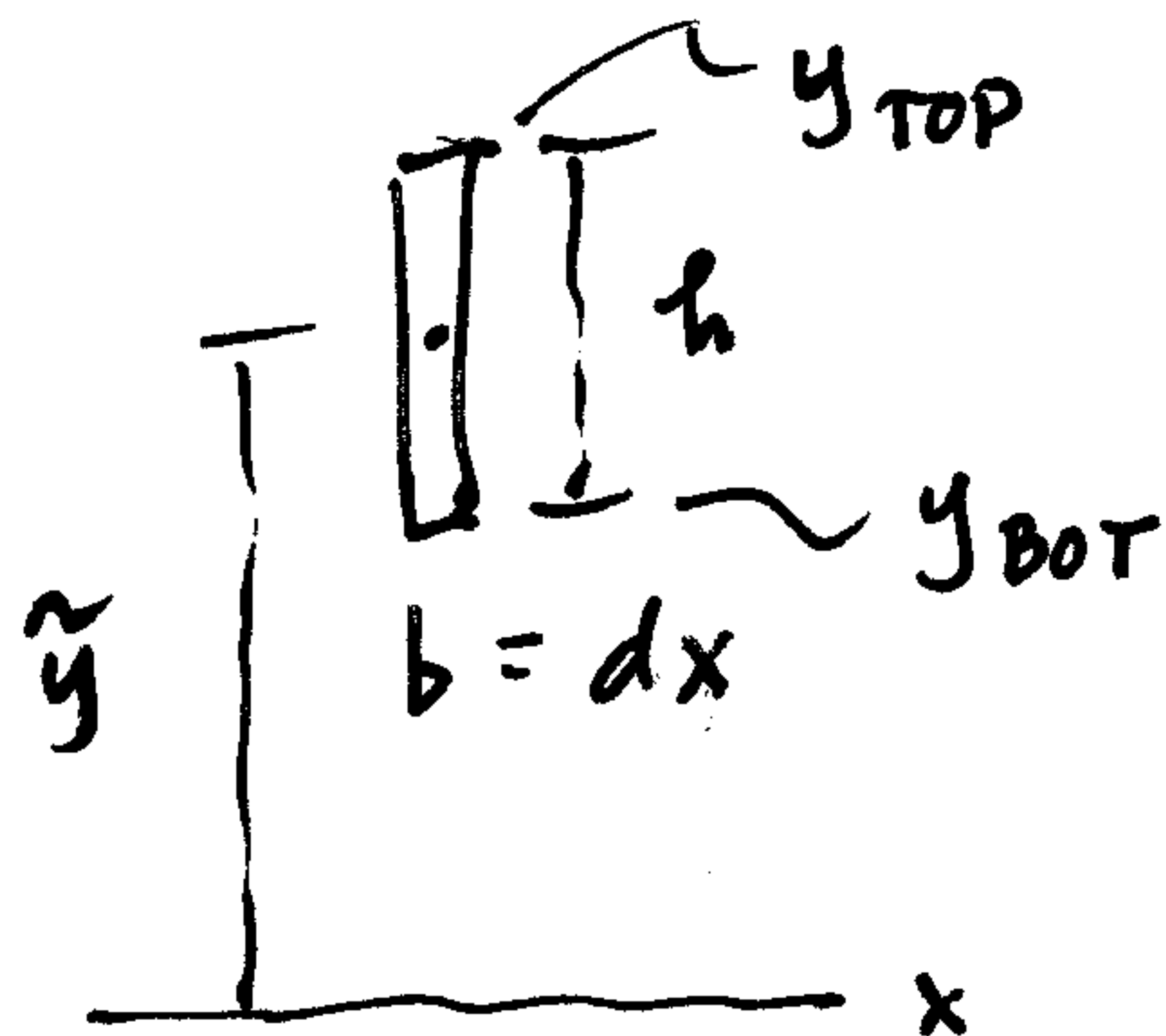
⑧ What about an element
 $\perp$ to the x axis?

⑥

We still write $I_x = \int dI_x$

BUT what is dI_x ?

Must use the PAT in differential form:



$$d\bar{I}_{x'} = \frac{1}{12} b h^3$$

$$= \frac{1}{12} (dx) (y_{TOP} - y_{BOT})^3 \quad !$$

$$\underline{\text{and}} \quad \tilde{y} = \left(\frac{y_{BOT} + y_{TOP}}{2} \right) \quad !$$

Then, $I_x = \int dI_x = \int (d\bar{I}_{x'} + dA \cdot \tilde{y}^2)$

$$I_x = \int \left[\frac{1}{12} (y_{TOP} - y_{BOT})^3 dx + \right.$$

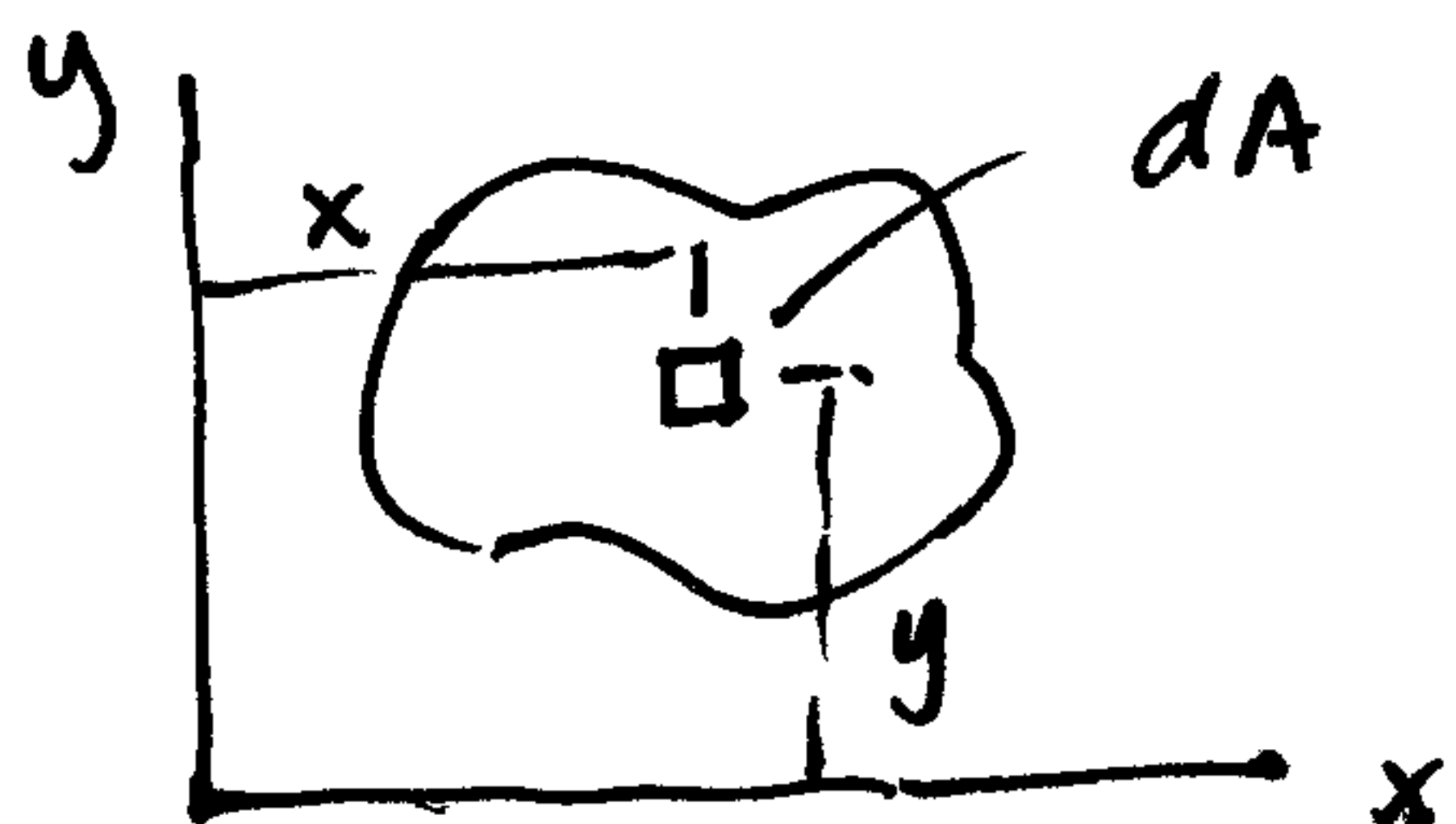
$$\left. \left[\frac{(y_{BOT} + y_{TOP})}{2} \right]^2 \cdot (y_{TOP} - y_{BOT}) dx \right]$$

this is dA

⑨ What about an element

$$dA = dx dy \text{ or } dA = dy dx ?$$

Use the PAT in differential form:



$$dI_x = d\bar{I}_{x'} + y^2 dA$$

$$\square \frac{dy}{dx}$$

$$d\bar{I}_{x'} = \frac{1}{12} b h^3$$

$$= \frac{1}{12} (dx) (dy)^3 \Rightarrow 0$$

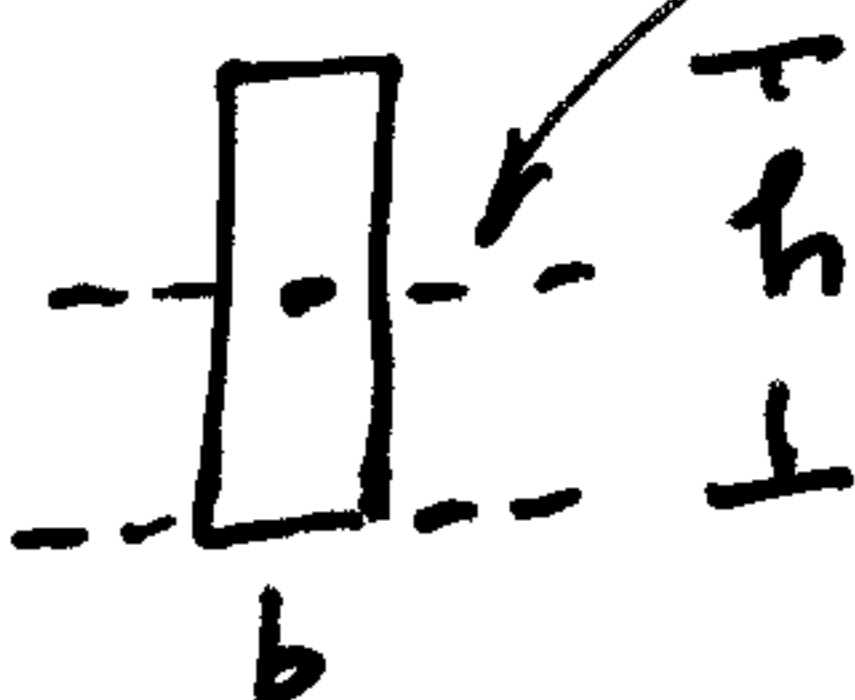
thus, $I_x = \int dI_x$

$$\boxed{I_x = \int y^2 dA}$$

as expected...

⑩ One other case: Can use a $\perp$ element if it touches the axis.

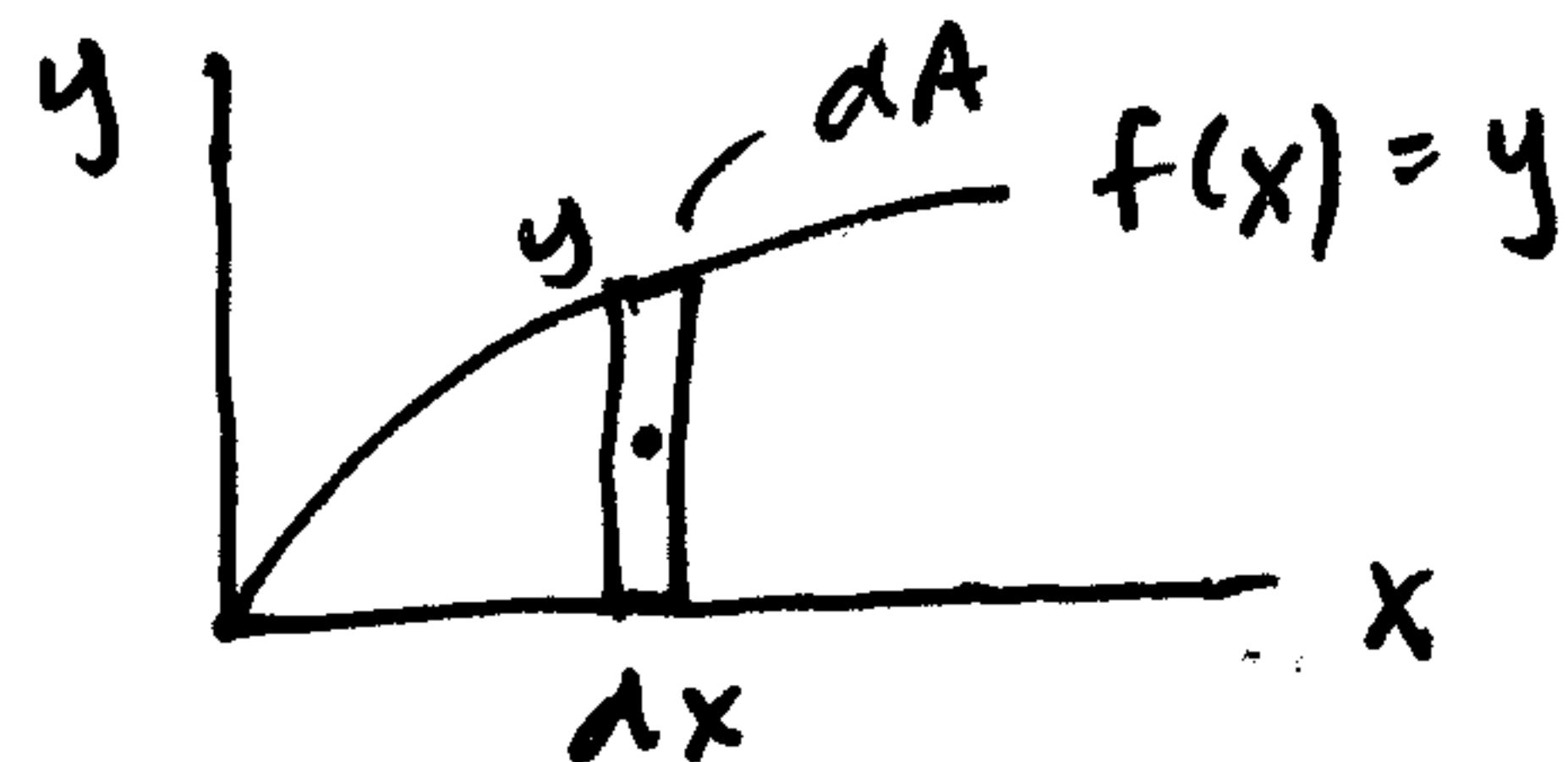
How? For a rectangle,



$$\boxed{\bar{I} = \frac{1}{12} b h^3}$$

$$\boxed{I_{base} = \frac{1}{3} b h^3}$$

So, if have an area which touches the x-axis



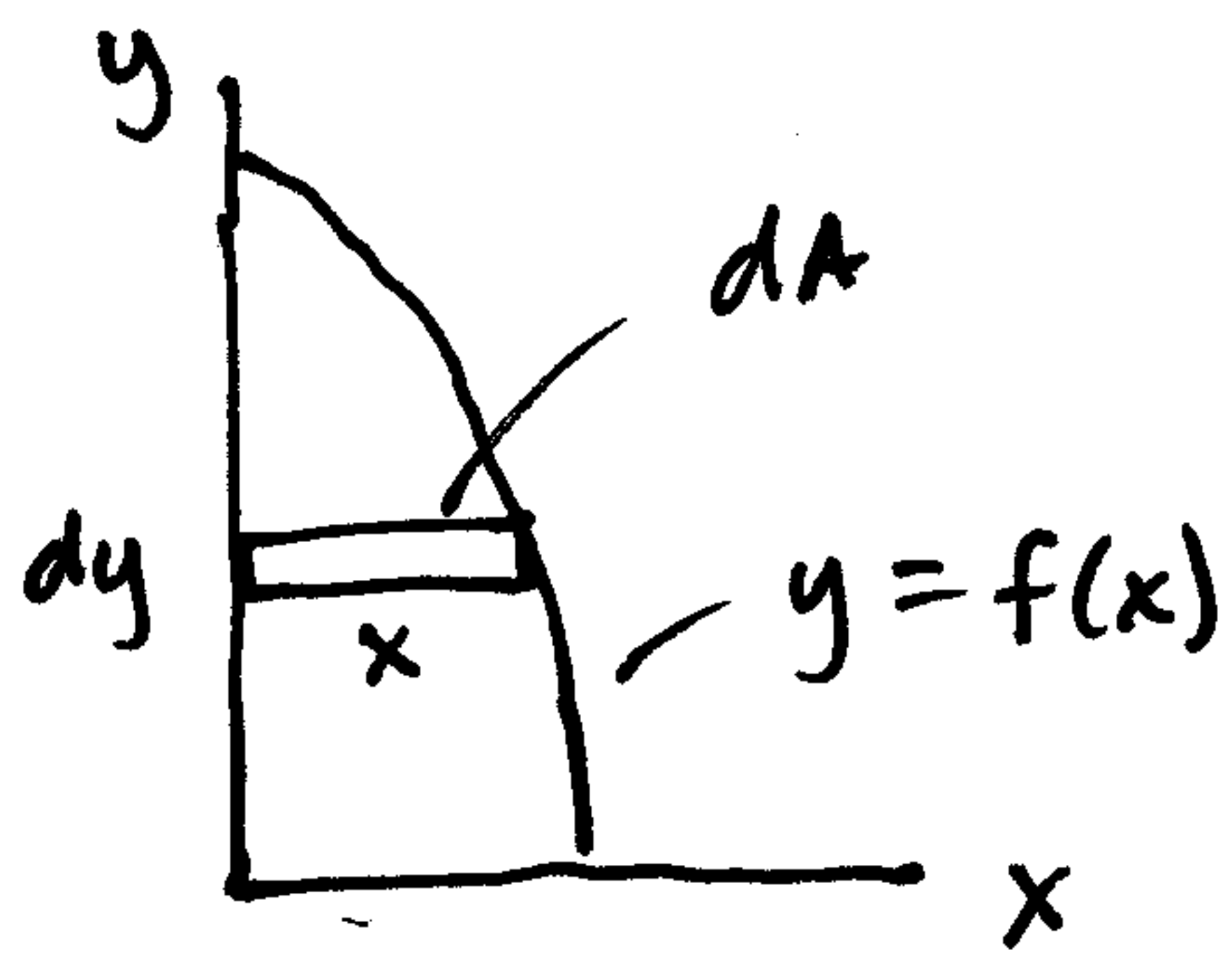
$$I_x = \int dI_x$$

BUT, $dI_x = \frac{1}{3} b h^3$

$$= \frac{1}{3} (dx) y^3 !$$

and $\boxed{I_x = \int \frac{1}{3} y^3 dx}$
 $\quad \quad \quad \uparrow$
 $\quad \quad \quad y = f(x)$

⑩ Similarly, for I_y , if area touches the y axis...



$$I_y = \int dI_y$$

$$\text{But } dI_y = \frac{1}{3} b h^3 \\ = \frac{1}{3} (dy) x^3$$

↑ convert to y function

Finally,

$$I_y = \int \frac{1}{3} x^3 dy$$

↑ convert to y function
